

STUDY PROTOCOL

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Comparison of a novel gamified binocular therapy and standard monocular patching therapy in treating unilateral amblyopia in young children: a prospective, multicenter, randomized controlled trial

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Abstract

Background Amblyopia is a leading cause of visual impairment in children, and conventional treatments often encounter challenges related to poor compliance and adverse psychosocial effects. Recent advancements in treatment approaches have evolved beyond merely suppressing the dominant eye, now focusing on improving binocular visual function. Binocular therapy has emerged as a promising and innovative approach. The study evaluates the efficacy of a novel gamified binocular therapy for treating unilateral amblyopia on comprehensive visual functions.

Methods This prospective, multicenter, randomized, blinded, controlled trial is designed as a non-inferiority study. A total of 144 patients, aged 4 to 7 years, diagnosed with amblyopia, will be randomly assigned to either the intervention group (72 patients) or the control group (72 patients). Participants in the intervention group will engage in daily 1-h sessions of gamified binocular therapy, 5 days per week, for a total duration of 8 weeks (40 h). The treatment employs a roguelike shooting game, presented through binocular presentation via two distinct channels, powered by a real-time artificial intelligence-driven visual engine. The control group will receive traditional patching therapy for 2 h per day, 7 days per week for 8 weeks (112 h). The primary outcome measure of this study is distance visual acuity, while secondary outcome measures include stereoacuity and developmental visual perception assessed using the Developmental Test of Visual Perception (DTVP). These measures will be assessed at baseline and follow-up visits to determine the effectiveness of the intervention.

Discussion This study is a multicenter randomized trial to evaluate the efficacy and safety of the Vision Planet Training System for the treatment of unilateral amblyopia in young children compared with standard monocular patching therapy. The results are anticipated to offer valuable insights into the efficacy of binocular treatment for unilateral amblyopia.

Trial registration Chinese Clinical Trial Registry ChiCTR2300079090. Registered on 25 December 2023. <https://www.chictr.org.cn/showproj.html?proj=215374>.

Keywords Amblyopia, Binocular therapy, Serious game, Visual acuity, Stereoacuity, Patching

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Administrative information

Note: the numbers in curly brackets in this protocol refer to SPIRIT checklist item numbers. The order of the items has been modified to group similar items (see <http://www.equator-network.org/reporting-guidelines/spirit-2013-statement-defining-standard-protocol-items-for-clinical-trials/>)

Title {1}	Comparison of a novel gamified binocular therapy and standard monocular patching therapy in treating unilateral amblyopia in young children: a prospective, multicenter, randomized controlled trial
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Role of sponsor {5c}	Rui Liu contributed to the study design, collection, management, and the decision to submit the report for publication.

All items from the WHO Trial Registration Data Set are available in the trial registration record (ChiCTR [ChiCTR2300079090]) and supplemented by details provided in this protocol.

Introduction

Background and rationale {6a}

Amblyopia, a leading cause of visual impairment in young children, has a substantial impact on their overall

quality of life. The prevalence of amblyopia in children aged 30–71 months ranges from 0.7 to 2.6%, while in older children it varies between 1.0 and 5.5% [1–3]. The risk factors for amblyopia include uncorrected refractive error, anisometropia, strabismus, form deprivation, or any combination of conditions [4]. Anisometropia is associated with an increased likelihood of developing amblyopia, observed in approximately 33% of children diagnosed with this condition [5]. Strabismus is linked to 19% to 50% of cases involving unilateral amblyopia, significantly increasing the risk of developing amblyopia by a factor ranging from 2.7 to 18 times [6]. Beyond visual impairment, amblyopia can lead to dysfunction in accommodation, fixation, binocularity, vergence, reading fluency, contrast sensitivity, motion perception, and excessive crowding [7–12].

The standard treatment for amblyopia, whether anisometropic, strabismic, or mixed-in origin, begins with refractive correction, irrespective of its underlying cause. This is typically followed by patching or atropine penalization of the dominant eye to encourage the use of the amblyopic eye [13–16]. Patching has been the conventional approach and can lead to significant improvements in visual acuity (often by one or two lines) for many patients [17–19]. However, adherence to patching is challenging, with roughly half of participants failing to follow the regimen due to its adverse impact on psychological and social well-being [20]. Additionally, patching does not effectively improve binocular vision and interferes with reading speed, motion processing, and visual perception [21–24].

In recent years, treatment strategies for amblyopia have shifted from monocular suppressing of the dominant eye to improving binocular visual function [25, 26]. This new focus acknowledges that many children with amblyopia retain some binocular vision and that the visual cortex can support vision in the weaker eye through reduced contrast sensitivity in the dominant eye. This understanding has led to the development of binocular vision therapies in clinical practice [27, 28].

Several studies have demonstrated that binocular therapy, often through interactive activities such as video games, movies, and shutter glasses, can improve visual acuity (VA) [29, 30]. For instance, a multicenter randomized controlled trial demonstrated that binocular digital treatment led to a 1.8-line enhancement in VA for the amblyopic eye in children aged 4–7 [31]. Similarly, another randomized controlled trial in children aged 4–9 reported a 2.8-line improvement after a 16-week binocular home treatment, which was comparable to those of patching therapy (2.3 lines improvement) [32]. Gamification has also been widely used to improve adherence to interventions for amblyopia and enhance

visual-perceptual defects in pediatric patients [33, 34]. One common form of gamification therapy is the falling blocks game, which has been shown to improve VA in the amblyopia eye by 1.05 lines in children aged 5 to 13 years, compared to a 1.35-line improvement in the patching group after 16 weeks [35]. Another widely used game, the Dig Rush game, demonstrated a 1.5-line enhancement in VA for children aged 4 to 10 years after 2 weeks of treatment, significantly outperforming the control group that received part-time patching (0.7-line improvement). Despite the promising visual outcomes observed in both the Falling Blocks and Dig Rush games, their ability to sustain long-term adherence remains a significant challenge. Studies have reported a decline in adherence over time, as the repetitive nature and limited engagement mechanics of these games fail to maintain patients' interest [36].

Amblyopia not only impairs visual acuity but also affects higher-order visual processing functions, such as visual-motor integration, spatial orientation, and figure-ground discrimination. These perceptual deficits can contribute to difficulties in academic performance, motor coordination, and everyday functional tasks. As a result, assessing the broader impact of amblyopia on visual perception is crucial to understanding the full extent of the condition and is not well investigated. One widely used tool for evaluating visual-perceptual skills in children is the Developmental Test of Visual Perception (DTVP), which assesses abilities like spatial relations and figure-ground discrimination. Given that amblyopia not only impairs visual acuity but also affects higher-order visual processing functions, the incorporation of DTVP in this study provides a more comprehensive understanding of the condition's impact. Assessing these perceptual deficits is crucial, as they have been associated with difficulties in academic performance, motor coordination, and daily functional tasks. Moreover, evaluating the efficacy of binocular digital therapy through objective measures like DTVP allows for a more thorough assessment of treatment outcomes beyond improvements in visual acuity. By integrating DTVP into the study protocol, this research aims to bridge the gap between clinical improvements and real-world functional performance, providing a more holistic perspective on amblyopia rehabilitation.

In this study, we evaluate the efficacy and safety of a novel binocular digital therapy in children with unilateral amblyopia through comprehensive assessments of visual function, especially binocular conditions for the first time. The Vision Planet System is a gamified medical software, incorporating rogue-like game mechanics (characterized by procedurally generated content and high re-playability) designed to treat amblyopia under binocular conditions. The system employs active viewing with

feedback, and these dynamic and interactive features have the potential to enhance patient motivation and promote long-term adherence. By reducing monotony in therapeutic exercises, the system aims to strengthen the amblyopic eye while minimizing participants' conscious awareness of the process.

Objectives {7}

The primary objective of this trial is to assess whether an 8-week gamified binocular therapy is comparable to standard monocular masking therapy in improving visual acuity and stereoacuity. Developmental visual perception and patient compliance and satisfaction will be evaluated. We hypothesize that, after 8 weeks of treatment, binocular digital therapy will demonstrate non-inferiority in distance VA and stereoacuity compared to part-time patching treatment.

Trial design {8}

The study is a prospective, multicenter, randomized control trial designed with a parallel-group, evaluator-blind, non-inferiority framework. Participants will be randomly allocated in a 1:1 ratio to either the intervention group, receiving gamified binocular therapy, or the control group, undergoing traditional monocular patching therapy. The study aims to determine whether the novel gamified binocular therapy is non-inferior to the standard patching approach in improving visual function in young children with unilateral amblyopia.

The study is conducted across three clinical centers in China and will follow an 8-week treatment period, with an optional crossover phase for participants. The primary outcome measure is distance visual acuity (VA) improvement, while secondary outcomes include stereoacuity, visual perception abilities (via the Developmental Test of Visual Perception, DTVP), and treatment adherence.

Methods: participants, interventions and outcomes

Study setting {9}

The study will adhere to the principles of the Declaration of Helsinki at three clinical sites in China, including the Eye & ENT Hospital of Fudan University, the Children's Hospital of Fudan University, and Shanghai Children's Medical Center. The study protocol and informed consent have been approved by the institutional review boards of the Eye & ENT Hospital of Fudan University.

These hospitals are leading ophthalmology and pediatric medical centers with well-established clinical research infrastructures and experienced medical teams specializing in pediatric ophthalmology and amblyopia treatment. The study sites have been carefully selected to ensure access to a diverse pediatric patient population, enabling comprehensive data collection and ensuring a high

standard of clinical and research expertise. All study procedures will adhere to the Declaration of Helsinki, and ethical approval has been obtained from the Institutional Review Board (IRB) of the Eye & ENT Hospital of Fudan University.

Eligibility criteria {10}

The inclusion and exclusion criteria are as follows:

Inclusion criteria

1. Age between 4 and 7 years;
2. Diagnosed with anisometropic, strabismic amblyopia, or combined conditions;
3. Strabismic amblyopia, defined as the presence of heterotropia at distance and/or near fixation, a history of strabismus surgery, or the resolution of strabismus following hyperopic spectacle correction, with a simultaneous prism cover test result of ≤ 5 prism diopters (PD);
4. Anisometropic amblyopia, defined as the presence of a spherical equivalent difference of ≥ 1.50 diopters (D) between the eyes or an astigmatic difference of ≥ 1.00 D in any meridian, with no associated strabismus;
5. Mixed-mechanism amblyopia, defined as amblyopia in the presence of both strabismus and anisometropia;
6. Amblyopic eye VA ranging from 0.2 LogMAR to 1.0 LogMAR;
7. Best-corrected visual acuity (BCVA) in the dominant eye of 0.3 LogMAR or better for children aged 4 to 5 years, and 0.2 LogMAR or better for children ≥ 5 years;
8. An interocular difference of ≥ 2 logMAR lines (≥ 10 letters);
9. Refractive error correction (based on cycloplegic refraction completed within the previous 6 months) if any of the following are true:

- Hypermetropia of ≥ 2.5 D by SE
- Astigmatism of ≥ 1.00 D
- Anisometropia of > 0.50 D

- a) Spectacle-prescribed instruction based on the cycloplegic refraction must have been completed within the previous 6 months:

- Spherical equivalent (SE) should not exceed ± 0.50 D when fully correcting anisometropia

- SE under-correction must not exceed 1.50 D SE, and the reduction in plus spherical power must be symmetrical for both eyes
- The cylinder power in both eyes should be adjusted to ensure it is within 0.50 D of fully correcting the astigmatism
- The axis must be within $+10^\circ$ if the cylinder power is ≤ 1.00 D and within $+5^\circ$ if the cylinder power exceeds 1.00 D

- b) Spectacle corrections that fulfill the aforementioned criteria should be strictly followed:

- For a minimum of 8 weeks or until distance VA stability is documented (defined as less than 0.1 logMAR change using the same testing method, measured on two consecutive examinations at least 8 weeks apart)

To assess the stability (non-improvement) of VA:

- The first of two VA measurements can be conducted under one of the following conditions: (1) while wearing the current spectacles, (2) using trial frames with or without cycloplegia, or (3) uncorrected if a new prescription has been provided
- The second measurement must be performed without cycloplegia, within corrected spectacles that have been worn for consistently at least 4 weeks

10. Consistent spectacle wear for a minimum of 8 weeks, as required;
11. A cessation of patching or any binocular therapy for a minimum duration of 2 weeks.

Exclusion criteria

1. Diagnosed as myopia;
2. History of intraocular surgery, refractive surgery, or any organic ocular disease associated with visual impairment, or any other cause potentially reducing best-corrected visual acuity (BCVA)
3. History of photosensitive epilepsy, use of rigid gas-permeable (RGP) contact lenses, or any reported ocular anatomical abnormalities (e.g., mild lens opacities, optic nerve fiber anomalies);
4. Inability to wear spectacles or comply with study procedures, as determined by the investigator;

- 5. Any condition that may interfere with the participant's ability to complete the daily training (20 min to 1 h) in front of a near-view display;
- 6. Heterophoria with a total near deviation of 10 PD or greater.

Who will take informed consent? {26a}

Informed consent will be obtained from the parents or legal guardians of all participants before enrollment in this study. Study coordinators will explain the trial objectives, procedures, potential benefits, and risks to the parents and legal guardians.

Those expressing interest will undergo an eligibility and baseline assessment conducted by study-certified examiners. Eligible participants will be provided with a copy of the informed consent to review. Enrollment in the study will proceed only after written informed consent is obtained.

Additional consent provisions for collection and use of participant data and biological specimens {26b}

There are currently no planned ancillary studies using the data collected in this trial. The informed consent form also does not include provisions for future data reuse.

Interventions

Explanation for the choice of comparators {6b}

The comparator selected for this study is standard monocular patching therapy, which remains the gold standard in

the treatment of unilateral amblyopia in children. Patching has been widely adopted due to its proven efficacy in improving visual acuity by one to two lines [5]. It operates by occluding the dominant eye, thereby forcing the use of the amblyopic eye, and stimulating its neural pathways.

However, patching therapy faces notable challenges, particularly regarding adherence. Many young children experience discomfort and psychosocial issues associated with wearing an eye patch, leading to low compliance rates [20]. Furthermore, patching focuses solely on monocular visual improvement, neglecting binocular vision, which may result in persistent stereoacuity deficits and poor binocular function [21–24].

Given these considerations, patching serves as an ideal comparator, enabling the evaluation of whether the novel gamified binocular therapy offers equivalent or superior improvements in visual acuity, binocular vision, and patient adherence.

Intervention description {11a}

Gamified binocular therapy (intervention group)

Participants in the intervention group will undergo therapy using the Vision Planet Visual Training System, a home-based, gamified system developed for amblyopia treatment. The system integrates a roguelike shooting game designed to improve binocular vision and patient engagement, as depicted in Fig. 1.

1) Equipment:

- An 11-inch HUAWEI MatePad (Model DBR-W00) delivering visual stimuli

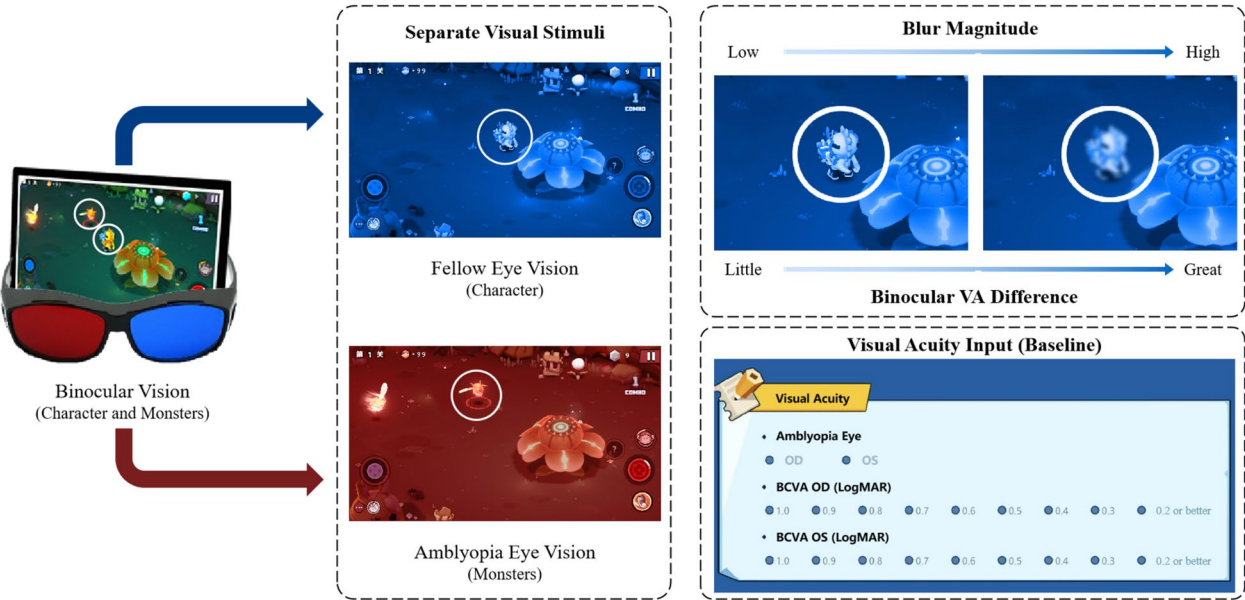


Fig. 1 Binocular treatment setup

- Anaglyph glasses for dichoptic image separation, applying Gaussian blur to the dominant eye

2) Therapy protocol:

- Duration: 8 weeks
- Frequency: 5 days per week
- Session length: 60 min per session (totaling 40 h)
- Mode: Home-based, parent-supervised therapy

The therapy dynamically adjusts contrast and blur based on the interocular acuity difference. Initial sessions focus on monocular tasks and gradually progress to binocular tasks and stereoacuity training as visual acuity improves. The interactive game mechanics encourage binocular fusion and visual-motor coordination, promoting visual rehabilitation.

1. Standard monocular patching therapy (control group):

The control group will follow conventional patching protocols:

- Duration: 8 weeks
- Frequency: 7 days per week
- Session length: 2 h per day (totaling 112 h)
- Mode: home-based, parent-supervised patching
Parents will receive guidance on proper patching techniques, ensuring complete occlusion of the dominant eye.

2. Cross-over option:

After 8 weeks, participants from both groups may cross over to the alternative treatment for an additional 8 weeks.

3. Monitoring:

Adherence and safety will be evaluated through scheduled follow-ups at 1, 4, 8, and 16 weeks, using standardized tests for visual acuity, stereoacuity, and visual perception.

Criteria for discontinuing or modifying allocated interventions {11b}

Participants may discontinue or modify their assigned interventions under the following conditions:

1) Adverse events:

- Development of diplopia, worsening heterotropia (> 10 prism diopters), or significant decline in BCVA (≥ 0.2 LogMAR)

- Onset of photosensitive epilepsy symptoms or any serious adverse events related to visual stimuli exposure

2) Participant request:

- Parents/legal guardians may request discontinuation or modification due to concerns over child discomfort, lack of engagement, or personal reasons.

3) Non-compliance:

- Persistent non-compliance, defined as completing less than 50% of prescribed sessions over two consecutive weeks, may prompt discontinuation or protocol modification after consultation.

4) Clinical judgment:

- Investigators may modify or terminate the intervention if deemed medically necessary based on participant responses or unforeseen risks.
- Discontinued participants will be encouraged to complete follow-up assessments for safety monitoring and data integrity.

Strategies to improve adherence to interventions {11c}

To optimize adherence to the intervention protocols, the following strategies will be implemented:

1. Gamification:

The Vision Planet system incorporates engaging game mechanics (e.g., levels, rewards, interactive feedback) to maintain children's motivation throughout the therapy.

2. Parental involvement:

Parents will receive comprehensive training on administering both therapies and monitoring adherence, including the use of parental logs.

3. Built-in monitoring:

The Vision Planet software tracks usage time and task completion, transmitting data to the research team for adherence monitoring.

4. Regular follow-ups:

Adherence assessments, parental interviews, and reinforcement of the treatment's importance will be conducted during scheduled visits at weeks 1, 4, 8, and 16.

Relevant concomitant care permitted or prohibited during the trial {11d}

1. Permitted concomitant care:

- 1) Spectacle usage:
 - All participants must wear their prescribed optical correction during therapy sessions and daily activities throughout the study.
- 2) Refractive error monitoring:
 - Routine ophthalmic check-ups, including refractive error assessments, are allowed during the study.
2. Prohibited concomitant care:
 - 1) Additional amblyopia treatments:
 - No participant may receive any form of amblyopia treatment outside the assigned intervention (e.g., vision therapy, atropine penalization) during the trial.
 - 2) New refractive corrections:
 - Changes to spectacle prescriptions are prohibited unless clinically necessary due to significant visual changes, in which case the participant will be reevaluated.
 - 3) Ocular medications:
 - Use of any topical ocular medications that may affect visual function (e.g., atropine) is not allowed during the trial period.

Provisions for post-trial care {30}

1. Post-trial care:

After completing the 16-week study period (including cross-over if applicable), participants will receive a final comprehensive ophthalmic assessment. Participants demonstrating continued amblyopia will be referred for further standard care, including patching, vision therapy, or other clinically appropriate interventions, at no cost for an additional 3 months post-trial.
2. Compensation for harm:

In the event of harm or adverse effects resulting from participation in the trial, medical care will be provided at the Eye & ENT Hospital of Fudan University without cost to the participant. The study adheres to institutional and national regulations concerning participant safety, and liability coverage has been arranged to ensure participants receive appropriate care in case of injury related to trial procedures.

Outcomes {12}

Primary outcome

The primary outcome is the change in best-corrected visual acuity (BCVA) in the amblyopic eye from baseline. Visual acuity will be measured using a logMAR

chart, either the Lea Symbols or ATS-HOTV chart, following the standardized protocol used in the Amblyopia Treatment Study (ATS) for children under 7 years of age. The metric is change from baseline, and the method of aggregation is mean change in BCVA at week 8 (primary endpoint), with secondary assessments at weeks 4 and 16. All testing will be conducted by certified masked examiners.

Secondary outcomes

The secondary outcomes of this trial include stereopsis, visual perceptual ability, and participant compliance and satisfaction.

Stereopsis, reflecting binocular depth perception, will be evaluated using the Randot Stereotest (Stereo Optical, USA) administered by a trained optometrist. Testing will be conducted at a near viewing distance of approximately 40 cm under habitual refractive correction. Stereoacuity scores will be recorded in arcseconds and log-transformed for statistical analysis. Participants who can only identify the stereoscopic fly or fail all stereotests will be assigned a conservative value of 3000 arcseconds. Between-group comparisons of stereoacuity at weeks 8 and 16 will be conducted using independent samples *t*-tests or Wilcoxon rank-sum tests, depending on the normality of the data distribution.

Visual perceptual ability will be evaluated using the Developmental Test of Visual Perception, Third Edition (DTVP-3), which includes subtests targeting motor-reduced visual perception and visual-motor integration. The primary analytic measure will be the change in the composite standardized score (General Visual Perception Quotient, GVPQ) from baseline to week 8. Group-level differences will be analyzed using parametric or nonparametric methods as appropriate.

Compliance and satisfaction represent secondary behavioral and subjective outcomes. Compliance will be quantified as the percentage of prescribed sessions completed over the 8-week treatment phase, automatically recorded via the digital system. Satisfaction will be assessed using a structured parent-reported questionnaire. In line with the crossover study design, satisfaction will be evaluated at week 8 and/or week 16, depending on the intervention sequence. All measures will be summarized using appropriate descriptive statistics and compared between groups using distribution-sensitive inferential tests.

Participant timeline {13}

The timeline for participant enrollment, interventions, assessments, and follow-up visits is structured to ensure consistency across all participants and sites. The trial follows a parallel design with an initial 8-week intervention

phase, an optional 8-week cross-over phase, and comprehensive follow-up assessments. The clinical trial schedule consists of several distinct phases designed to assess the efficacy and safety of the interventions. The screening and baseline phase (week -1 to 0) includes an eligibility screening, obtaining informed consent, and conducting baseline assessments, encompassing distance visual acuity, stereoacuity, the Developmental Test of Visual Perception (DTVP), and ocular alignment. Following baseline assessments, randomization (week 0) will be performed using stratified block randomization to

allocate participants into either the intervention or control group (see Fig. 2).

The intervention phase (week 1 to 8) involves two groups: the intervention group receives gamified binocular therapy administered 5 days per week for 60 min per session, while the control group undergoes standard monocular patching therapy for 7 days per week, 2 h daily. Throughout this phase, follow-up assessments are conducted at key intervals: week 1 focuses on adherence checks and safety monitoring; week 4 involves interim assessments of visual acuity, stereoacuity, and DTVP; and

	STUDY PERIOD					
	Enrolment	Allocation	Post-allocation			Close-out
TIMEPOINT	-1 week	0 week (Baseline)	1 week (Remote)	4-week	8-week (End of Treatment)	16-week (End of Follow-up)
ENROLMENT						
Eligibility screen	×					
Informed consent	×					
Allocation		×				
Demography		×				
INTERVENTIONS						
Gamified Binocular Therapy		←	→		←	→
Patching Therapy		←	→		←	→
ASSESSMENTS						
Distance Visual Acuity		×		×	×	×
Suppression		×		×	×	×
Strabismic Deviation		×		×	×	×
Near Stereopsis		×		×	×	×
Developmental Test of Visual Perception (DTVP)		×		×	×	×
Near Accommodative Amplitude (Optional)		×		×	×	×
Contrast Sensitivity (Optional)		×		×	×	×
Reading Test – DEM (Optional)		×		×	×	×
Safety Evaluation and Equipment Defects and Medication Records				×	×	×

Fig. 2 The schedule of enrollment, interventions and assessments

week 8 serves as the primary outcome assessment, evaluating distance best-corrected visual acuity (BCVA), stereoacuity, DTVP, treatment compliance, and any adverse events.

A cross-over option (week 9 to 16) will be offered, allowing participants to switch to the alternative therapy for an additional 8 weeks. Follow-up assessments will be conducted at weeks 12 and 16 to monitor progress. The study concludes with a final follow-up (week 16), which includes a comprehensive assessment of BCVA, stereoacuity, DTVP, compliance, and adverse events, followed by post-trial care planning. This structured approach ensures a thorough evaluation of both therapeutic interventions over the study period.

Sample size {14}

The required sample size was determined based on a non-inferiority study design. Assuming a one-sided significance level of 0.025, a power of 80%, a pooled standard deviation of 0.12 [32, 36], and a non-inferiority margin of 0.1 logMAR [37], calculations were performed using PASS software version 15.0.5. The required sample size was 64 per group to achieve the desired power; the study aims to recruit 144 participants (72 per arm) to account for an anticipated dropout rate of 10%.

Recruitment {15}

Achieving the target sample size of 144 participants (72 per arm) is critical for ensuring the statistical power of this non-inferiority trial. The following recruitment strategies will be implemented using a multi-center recruitment approach. Participants will be recruited from three high-volume ophthalmology centers in China: Eye & ENT Hospital of Fudan University, Children's Hospital of Fudan University, and Shanghai Children's Medical Center. These centers collectively see many pediatric patients with amblyopia, increasing the potential participant pool.

Assignment of interventions: allocation

Sequence generation {16a}

The allocation sequence for this trial will be generated using a computer-based randomization method to ensure an unbiased distribution of participants across the intervention groups. Specifically, the SAS 9.4 PLAN procedure will be utilized to create the randomization sequence. Participants will be randomly assigned to either the intervention or control group at a 1:1 ratio. Randomization will be stratified based on key factors, including the study center—Eye & ENT Hospital of Fudan University, Children's Hospital of Fudan University, and Shanghai Children's Medical Center—and the type of amblyopia (anisometropic, strabismic, or mixed).

To maintain balance across treatment arms, stratified block randomization will be implemented within each study center. Details of block sizes and the randomization sequence will be kept confidential in a secure, password-protected document and will not be accessible to investigators responsible for participant enrollment or assignment. This approach ensures allocation and upholds the integrity of the randomization process (see Fig. 3).

Concealment mechanism {16b}

To maintain allocation concealment and prevent selection bias, the following procedures will be used:

1. Centralized randomization system:

The allocation sequence will be managed centrally by an independent statistician who is not involved in participant enrollment or clinical assessments. Investigators enrolling participants will contact the central randomization office via a secure, password-protected web-based system to obtain the treatment allocation after confirming participant eligibility.

2. Opaque sealed envelopes (backup method):

In case of technical issues with the central system, sequentially numbered, opaque, sealed envelopes containing the group assignments will be prepared by the independent statistician.

Each envelope will only be opened by the study coordinator after confirming participant eligibility and obtaining informed consent. This approach ensures that the allocation sequence remains concealed until the point of intervention assignment, thereby preserving the integrity of the randomization process.

Implementation {16c}

1. Sequence generation

An independent statistician, not involved in participant enrollment or data analysis, will generate the allocation sequence using the SAS 9.4 PLAN procedure.

2. Participant enrollment

Study coordinators at each site will be responsible for enrolling participants. They will conduct eligibility screening, obtain informed consent, and perform baseline assessments before randomization.

3. Assignment to interventions

After enrollment and baseline assessments, the study coordinators will access the central randomization system (or use the sealed envelope method if needed) to determine the assigned intervention. Study-certified examiners responsible for outcome assessments

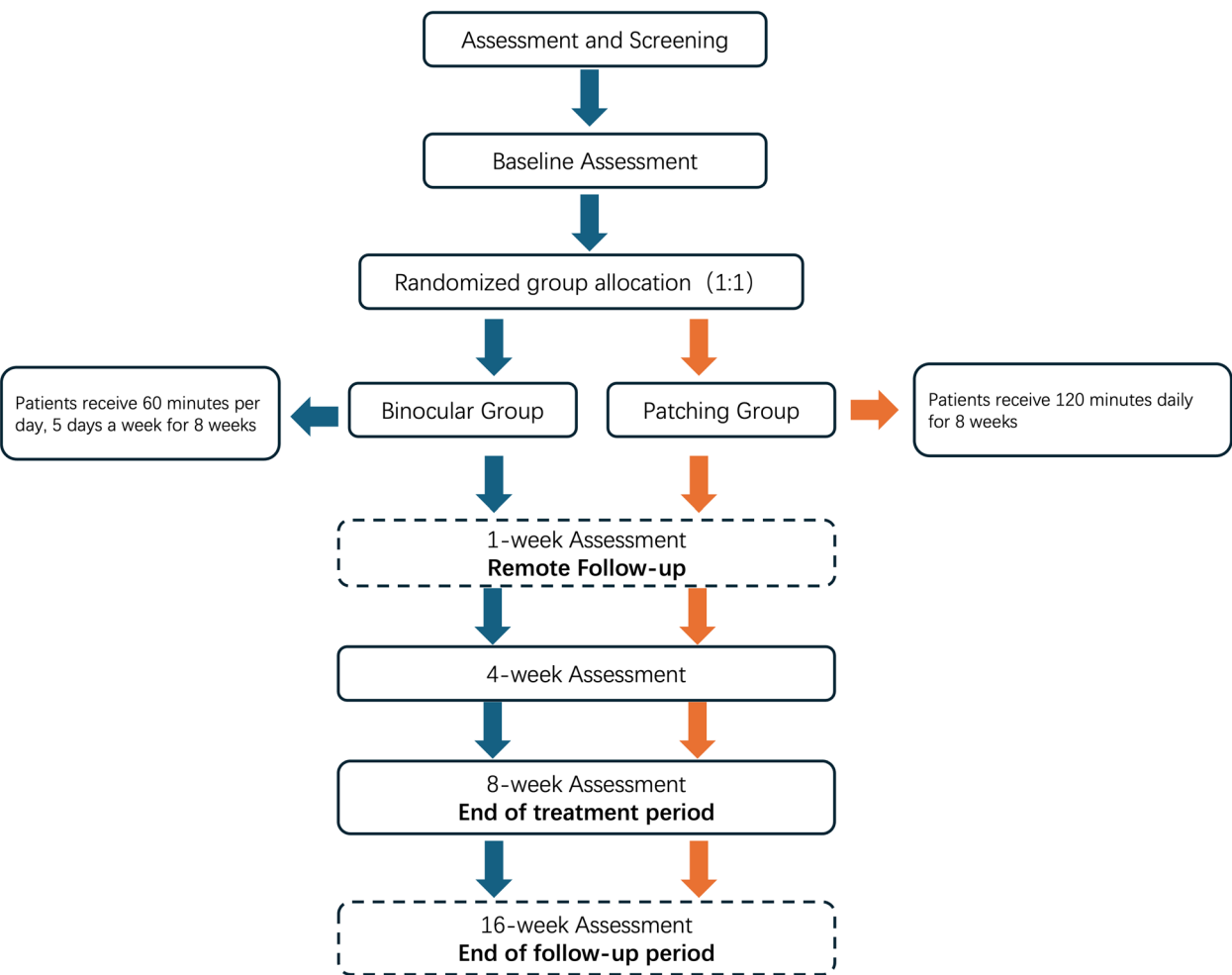


Fig. 3 Flowchart of the study

will remain blinded to the group allocation to minimize assessment bias.

Assignment of interventions: blinding

Who will be blinded {17a}

This trial employs a single-blind design to minimize assessment bias while recognizing the practical challenges of blinding in behavioral interventions. In this design, specific parties will be blinded to group allocations to preserve the integrity of outcome assessments.

Blinded parties include outcome assessors and data analysts. Study-certified examiners are responsible for measuring primary and secondary outcomes, such as best-corrected visual acuity (BCVA), stereoacuity, and Developmental Test of Visual Perception (DTVP) scores, which will be blinded to group assignments. These assessors have no access to randomization data or information regarding participants’assigned

interventions. To maintain this blinding, participants and their caregivers will be instructed not to disclose their assigned treatment during assessments.

Data analysts conducting statistical analyses will be also blinded to treatment allocation. Independent data analysts perform all analyses, with intervention groups coded (e.g., group A and group B) to ensure that blinding will be maintained throughout the data analysis process.

Procedure for unblinding if needed {17b}

While the trial is designed to maintain blinding, specific circumstances necessitate unblinding to ensure participant safety or address critical clinical concerns. Unblinding is permitted under the following conditions:

- Serious adverse events (SAEs): If a participant experiences an SAE potentially related to the intervention—such as significant visual deterioration, diplo-

pia, or severe ocular symptoms—unblinding will be conducted to facilitate appropriate clinical management.

- Participant withdrawal: If a participant withdraws from the trial due to adverse events or protocol concerns, unblinding can occur to aid in post-trial care and ensure continuity of treatment.
- The unblinding procedure follows a structured process to maintain the integrity of the trial:
- DSMC Review: The DSMC evaluates the request and determines whether unblinding will be justified based on participant safety and clinical necessity.

Unblinding execution: Upon approval, the independent statistician—who holds the randomization key—reveals the participant's assigned intervention directly to the site investigator. Importantly, outcome assessors and data analysts remain blinded unless unblinding is deemed essential for safety monitoring or data integrity. Documentation and reporting of all unblinding events will be meticulously maintained. Each event will be thoroughly documented, including the reason for unblinding, the date of the event, and the personnel involved.

Data collection and management

Plans for assessment and collection of outcomes {18a}

All outcome data, including baseline and follow-up measures, will be collected using standardized procedures conducted by study-certified examiners who will be blinded to treatment allocation. To ensure consistency and reliability across all study sites, assessors undergo comprehensive training in data collection protocols. The primary outcome is Distance Best-Corrected Visual Acuity (BCVA), assessed using an electronic visual acuity system following the Pediatric Eye Disease Investigator Group (PEDIG) Amblyopia Treatment Study protocols. LEA symbols with crowded optotypes will be utilized, and visual acuity will be recorded in logMAR units.

Secondary outcomes include stereoacuity, measured using the Titmus Stereo Acuity Chart (Stereo Optical Co., US) under appropriate optical correction. The Developmental Test of Visual Perception (DTVP) will be used to evaluate higher-order visual processing, focusing on motor-reduced visual perception (MRVP) and visual-motor integration (VMI). The DTVP is a validated and reliable tool for pediatric populations.

Compliance and satisfaction will be assessed using the Vision Planet Questionnaire, a validated instrument designed to evaluate treatment adherence, user experience, and overall satisfaction with the therapy. Adverse events monitoring will be systematically performed throughout follow-up visits, documenting events such

as diplopia, worsening heterotropia, and BCVA deterioration. This rigorous assessment framework ensures comprehensive data collection for evaluating treatment efficacy and safety.

Each clinical site will have a designated principal investigator (PI), a study coordinator, and at least one study-certified examiner. The PIs will be responsible for ensuring the accuracy, integrity, and timeliness of the data. To ensure consistency across sites, comprehensive training on case report forms (CRFs) will be provided before the study's initiation.

Plans to promote participant retention and complete follow-up {18b}

Maximizing participant retention and ensuring complete follow-up is critical for maintaining the integrity of the study. To promote retention, several strategies were implemented. Flexible scheduling allows follow-up visits to be arranged at times convenient for families, minimizing missed appointments. Besides, automated reminders via phone calls, text messages, or emails will be sent before each scheduled visit.

Data management {19}

Upon enrollment, CRFs will record participant data, and accuracy will be verified through double-checking. Any modifications to the CRFs must be documented with clear traceability and explanations. All study-related documents, including source documents, informed consent forms, and CRFs, will be securely stored in a designated cabinet. The Data and Safety Monitoring Committee of the Eye & ENT Hospital of Fudan University will oversee data and safety monitoring and the overall conduct of the study. All study documents and data will be securely archived for a minimum of 10 years following the end of the trial, in accordance with institutional and national regulations.

Confidentiality {27}

Participant confidentiality will be strictly maintained throughout the trial in compliance with national and institutional regulations. Data collection and storage procedures will be designed to safeguard personal information by storing personal identifiers (e.g., names and contact details) separately from clinical data. Each participant will be assigned a unique study identification number, which will be used consistently across all study documents and databases to ensure de-identification.

Data sharing and access will be restricted to authorized study personnel. Only designated staff has access to identifiable participant data, while outcome assessors and data analysts work exclusively with de-identified datasets to maintain confidentiality during data analysis.

Comprehensive confidentiality measures will be implemented to protect participant information. All electronic data will be encrypted during both storage and transmission to prevent unauthorized access. Physical documents, such as consent forms, will be securely stored in locked cabinets located in restricted areas. These protocols ensure the integrity and confidentiality of participant data throughout the study.

Plans for collection, laboratory evaluation and storage of biological specimens for genetic or molecular analysis in this trial/future use {33}

No biological specimens will be collected in this trial/future use.

Statistical methods

Statistical methods for primary and secondary outcomes {20a}

All analyses will follow the modified intention-to-treat approach. Continuous data will be presented as mean $\pm$ standard deviation if they conform to a normal distribution. Inter-group comparisons will be conducted using either an independent t-test or Wilcoxon rank-sum test, while a paired t-test or Wilcoxon signed-rank test will be used to compare pre- and post-treatment measurements. Dichotomous data will be reported as frequencies or percentages, and within-group comparisons will be analyzed using the chi-squared test or Fisher's exact test. Statistical analyses will be performed using SPSS software (version 20.0, IBM, Chicago, IL, USA), with a *P*-value of less than 0.025 considered statistically significant.

Interim analyses {21b}

One interim analysis will be conducted after 50% of participants (*n* = 72) complete the 8-week intervention phase to assess both efficacy and safety.

Methods for additional analyses (e.g., subgroup analyses) {20b}

Not applicable.

Methods in analysis to handle protocol non-adherence and any statistical methods to handle missing data {20c}

As outlined in the exclusion criteria, samples that complete less than 50% of the prescribed sessions over two consecutive weeks may be discontinued or subject to protocol modifications following consultation. To

maintain statistical power, an additional 10% of samples will be recruited for the study. Participants meeting the following criteria will be excluded from the analysis: (1) those who never received treatment; (2) those randomized but later deemed ineligible; and (3) those with no post-baseline data. Other missing data not covered by these criteria will be imputed using the last observation carried forward method.

Plans to give access to the full protocol, participant-level data and statistical code {31c}

Upon publication of the primary results, the study will ensure transparency and promote scientific collaboration by providing public access to key materials. The following resources will be made available to the scientific community: the full study protocol, a de-identified participant-level dataset, and the statistical code used for data analysis.

The data-sharing process will be managed to maintain participant confidentiality and data integrity. Researchers interested in accessing the protocol, dataset, or statistical code must contact the corresponding author to submit a formal request. All shared data will be fully de-identified to protect participant privacy. Additionally, researchers will be required to complete a data use agreement, which will outline the intended analyses and detailed plans for secure data handling.

All ethical considerations surrounding data sharing will strictly adhere to institutional and national guidelines on participant privacy and data protection, ensuring that data dissemination aligns with established ethical standards. This approach facilitates transparency while safeguarding participant confidentiality.

Oversight and monitoring

Composition of the coordinating centre and trial steering committee {5d}

The trial was centrally coordinated at the Eye & ENT Hospital of Fudan University, which served as the main coordinating center. The Coordinating Centre will be responsible for the day-to-day management of the trial, including overseeing trial logistics such as data collection, communication among study sites, and participant follow-up. The center ensures strict adherence to the study protocol and ethical guidelines, provides comprehensive training to site investigators and coordinators, and manages all regulatory submissions and approvals. To maintain oversight and address operational challenges, the coordinating center team convenes biweekly to review trial progress and implement necessary adjustments.

The Trial Steering Committee (TSC) provides overall supervision of the study, ensuring that it adheres to Good Clinical Practice (GCP) standards and ethical requirements. The TSC's responsibilities include monitoring trial conduct and progress, periodically reviewing safety and efficacy data, approving major protocol amendments, and offering guidance on scientific and ethical issues. The TSC meets quarterly throughout the study, with the flexibility to convene more frequently if required to address urgent matters.

Composition of the data monitoring committee, its role and reporting structure {21a}

DMC is not implemented in this study. The data quality is ensured by real-time validation of the inputs to the digital system designed by the coordinating committee, and rigid analysis will be performed to confirm the data quality.

Adverse event reporting and harms {22}

Adverse event (AE) monitoring will be conducted systematically throughout the trial to ensure participant safety. AEs will be collected at each scheduled follow-up visit as well as through spontaneous reports from participants or their caregivers. All AEs—whether solicited or unsolicited—will be documented in the case report forms (CRFs) using standardized ophthalmologic terminology, in accordance with protocol-defined procedures.

Investigators will evaluate each event for severity (mild, moderate, or severe), outcome, and its potential relationship to the study intervention. Specific attention will be paid to predefined anticipated risks, such as visual fatigue, headache, or visual acuity decline of ≥ 2 lines.

In the event that an AE necessitates discontinuation or modification of the assigned intervention, decisions will be guided by predefined safety criteria to ensure participant well-being while preserving the integrity of the study. Serious adverse events (SAEs), if any, will be reported within 24 h to the ethics committee and relevant regulatory authorities, in compliance with national safety-reporting requirements.

Frequency and plans for auditing trial conduct {23}

Independent audits will be conducted throughout the trial to ensure compliance with Good Clinical Practice (GCP) standards and strict adherence to the study protocol. These audits will be performed biannually at each clinical site by an independent auditing team that is not involved in the trial's conduct or data analysis, thereby maintaining objectivity and minimizing potential conflicts of interest.

Plans for communicating important protocol amendments to relevant parties (e.g. trial participants, ethical committees) {25}

Significant protocol modifications, including changes to eligibility criteria, intervention protocols, or outcome measures, will be promptly communicated to all relevant stakeholders to ensure transparency and regulatory compliance. The communication plan for protocol amendments includes several key components. First, any major protocol changes will be submitted to Ethical Committees (IRBs/RECs) for review and approval before implementation, ensuring that ethical standards are maintained. Amendments that may impact participant safety, data integrity, or the overall conduct of the trial will be reported to Regulatory Authorities following applicable guidelines.

To maintain transparency, updates will be made to the trial's registration on the Chinese Clinical Trial Registry (ChiCTR2300079090) to reflect any protocol amendments. Investigators and study sites will be informed of changes through formal communications and, if necessary, additional training sessions will be organized to ensure proper implementation of the amendments. Trial participants will be notified of any changes that could affect their participation, with revised informed consent forms provided for significant amendments to ensure continued informed participation.

Additionally, any protocol changes impacting study design or outcomes will be reported to the scientific community through future publications or study reports. All protocol amendments and related communications will be thoroughly documented and archived for audit and regulatory purposes, ensuring full traceability and compliance with applicable standards.

Dissemination plans {31a}

For Healthcare Professionals, results will be shared with ophthalmologists, pediatricians, and optometrists through presentations at relevant national and international conferences. Educational seminars and workshops will be organized for clinicians involved in amblyopia management to translate research findings into clinical practice. For the scientific community, the trial results will be submitted for publication in peer-reviewed journals specializing in ophthalmology and pediatric care. Findings will be reported following the CONSORT guideline to ensure comprehensive and standardized reporting. For Trial Registries and Databases, results will be submitted to the Chinese Clinical Trial Registry (ChiCTR2300079090) within 12 months of trial completion, as per regulatory requirements. De-identified

data and findings will also be made available on public research databases, promoting data sharing and secondary analyses.

Discussion

This prospective, multicenter randomized controlled trial (RCT) aims to assess the efficacy and safety of Vision Planet binocular therapy for treating unilateral amblyopia in children aged 4 to 7 years, extending our previous work, leveraging well-established findings and supported by comprehensive data [38]. The study will be the first to assess the impact of this therapy on visual acuity, binocular visual function, and visual perception, providing a comprehensive understanding of its therapeutic potential. The system adheres to the principles of binocular treatment, which promote the use of both eyes simultaneously, enhancing cooperation between the eyes while targeting the amblyopic eye for improvement.

A key feature of the Vision Planet system is its gamified approach, which incorporates interactive game elements such as rewards, levels, and feedback to maintain children's engagement and improve treatment adherence. Poor adherence is a significant challenge in conventional therapies like patching, but the fun and engaging nature of gamification helps mitigate this issue, promoting consistent participation. This approach aims not only to improve visual acuity but also to foster better binocular vision, which is crucial for overall visual development.

The trial will also assess the therapy's impact on visual perception, including skills like spatial relations, figure-ground discrimination, and visual-motor integration. These perceptual abilities are important for daily functioning and academic success, making it essential to evaluate them alongside visual acuity. By comparing the effects of the Vision Planet system with a control group, the study will provide valuable data on its safety, feasibility, and effectiveness, contributing to the future of digital health interventions in pediatric eye care. Regardless of the outcomes, this trial will offer important insights into the potential of gamified therapies for treating amblyopia.

The study has several methodological strengths. First, it adopts a randomized controlled trial (RCT) design with a relatively large sample size, which provides sufficient statistical power to detect clinically meaningful differences and enhances the robustness of the findings. Second, the trial is conducted across three tertiary-level medical centers, improving the generalizability and representativeness of the sample. Finally, by incorporating the Developmental Test of Visual Perception (DTVP), the study extends outcome evaluation beyond visual acuity to include visual-perceptual functions, offering a more holistic understanding of treatment efficacy. However, a

key limitation is that the trial's follow-up duration is limited to 16 weeks. This timeframe may be insufficient to evaluate long-term visual outcomes or sustained compliance. Longer-term follow-up studies are needed to assess the durability of the observed effects.

Trial status

Study protocol version 1.0, dated April 24, 2023. Recruitment began on March 1, 2024. As of June 4, 2025, 71 participants have been enrolled, and 56 of them have completed the 16-week follow-up. Recruitment is ongoing and is expected to be completed by March 1, 2026.

Abbreviations

BCVA	Best-corrected visual acuity
CRFs	Case report forms
D	Diopter
DTVP	Developmental test of visual Perception
logMAR	Logarithm of the minimum angle of resolution
LEA	The LEA Symbols Chart
MRVP	Motor-reduced visual perception
PD	Prism diopters
PI	Principal investigator
RGP	Rigid gas-permeable
SE	Spherical equivalent
VA	Visual acuity
VMI	Visual-motor integration

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Authors' contributions {31b}

WQZ and JWZ were responsible for study conceptualization, methodology, and drafting and editing the manuscript. HYZ and JJZ were responsible for the software and methodology. JL and CLH were responsible for the recruitment of the participants and data collection. RL was responsible for study conceptualization, methodology, reviews, and correspondence. All authors read and approved the final manuscript.

Authorship for future trial-related publications will be determined based on the International Committee of Medical Journal Editors (ICMJE) criteria. The final order of authorship will reflect each contributor's actual and substantial contributions to study design, data acquisition, analysis, interpretation, and manuscript writing, as assessed following completion of the trial.

Funding {4}

This study was partially funded by BOKE Medical Technology (Shanghai) Co., Ltd., which provided financial support for software development, certain aspects of data coordination, and operational resources. However, data coordination was performed independently by the Eye and ENT Hospital of Fudan University to ensure data integrity and objectivity.

Data availability {29}

The data that support the findings of this study are available on request from the corresponding author, RL, upon reasonable request.

Declarations

Ethics approval and consent to participate {24}

The trial has been approved by the Ethical Review Board of the Eye & ENT Hospital of Fudan University (2023180). Written, informed consent to participate will be obtained from all participants.

Consent for publication {32}

Not applicable.

Competing interests [28]

Heying Zhang and Jianjuan Zhao are affiliated with Boke Medical (Shanghai) Co., Ltd., and contributed to the software and methodology. No other competing interests were declared.

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